

GPU-Accelerated Nonlinear Parameter Search

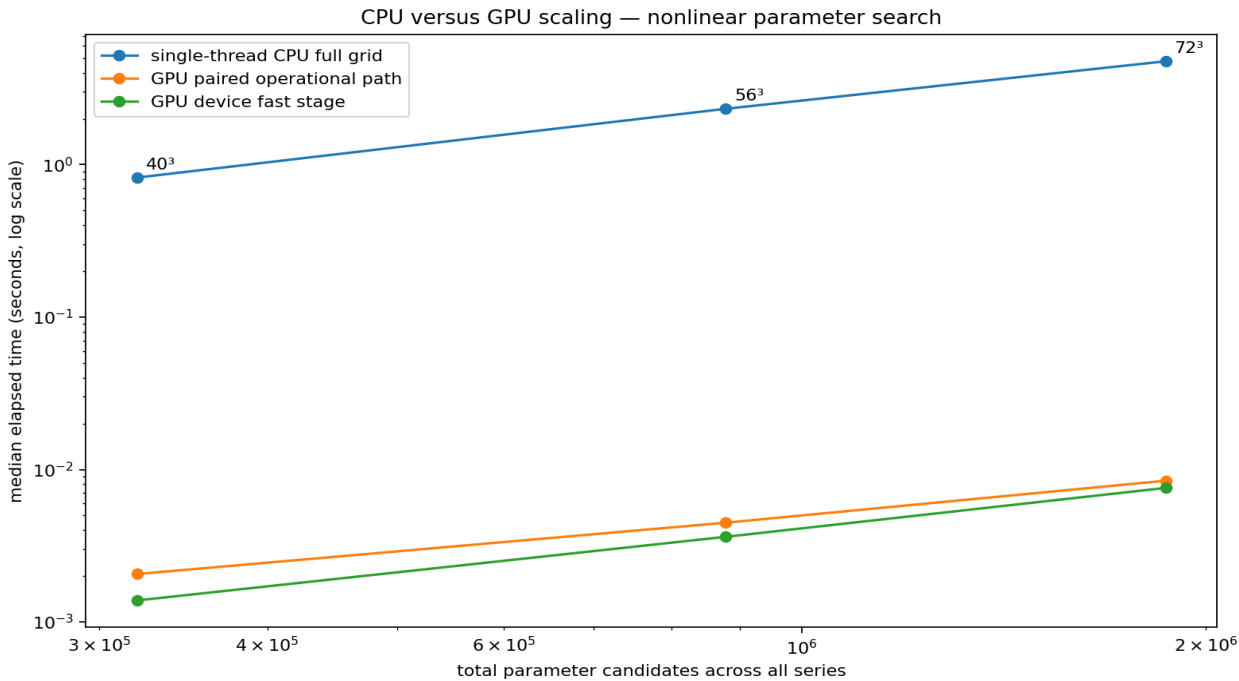
Neutral CUDA portfolio case study - full-grid screening, segmented Top-K selection, and refined validation.

Grid	Candidates	CPU median	GPU paired path	Speedup
40 ³	320,000	0.822287 s	0.002063 s	398.7x
56 ³	878,080	2.323822 s	0.004478 s	518.9x
72 ³	1,866,240	4.763587 s	0.008444 s	564.1x

Benchmark scope. The paired GPU operational path includes host-to-device transfer, GPU grid evaluation, segmented Top-K selection, and Top-K transfer back to the host. It excludes one-time allocation, CPU refinement, CSV writing, and figure rendering.

Validation. 25 GPU repetitions and 5 CPU repetitions after warm-up; medians reported. The maximum absolute difference between CPU and GPU best-grid MSE was 5.06e-09.

Why Top-K refinement. Across five deterministic synthetic series, refinement reduced MSE by 15.6% on average. One final winner came from fast rank 1, illustrating why retaining a shortlist is safer than trusting one grid point.



Hardware: NVIDIA GeForce GTX 1660 Ti (compute capability 7.5). Performance depends on workload, data layout, compiler, and hardware; this is not a universal speed guarantee.